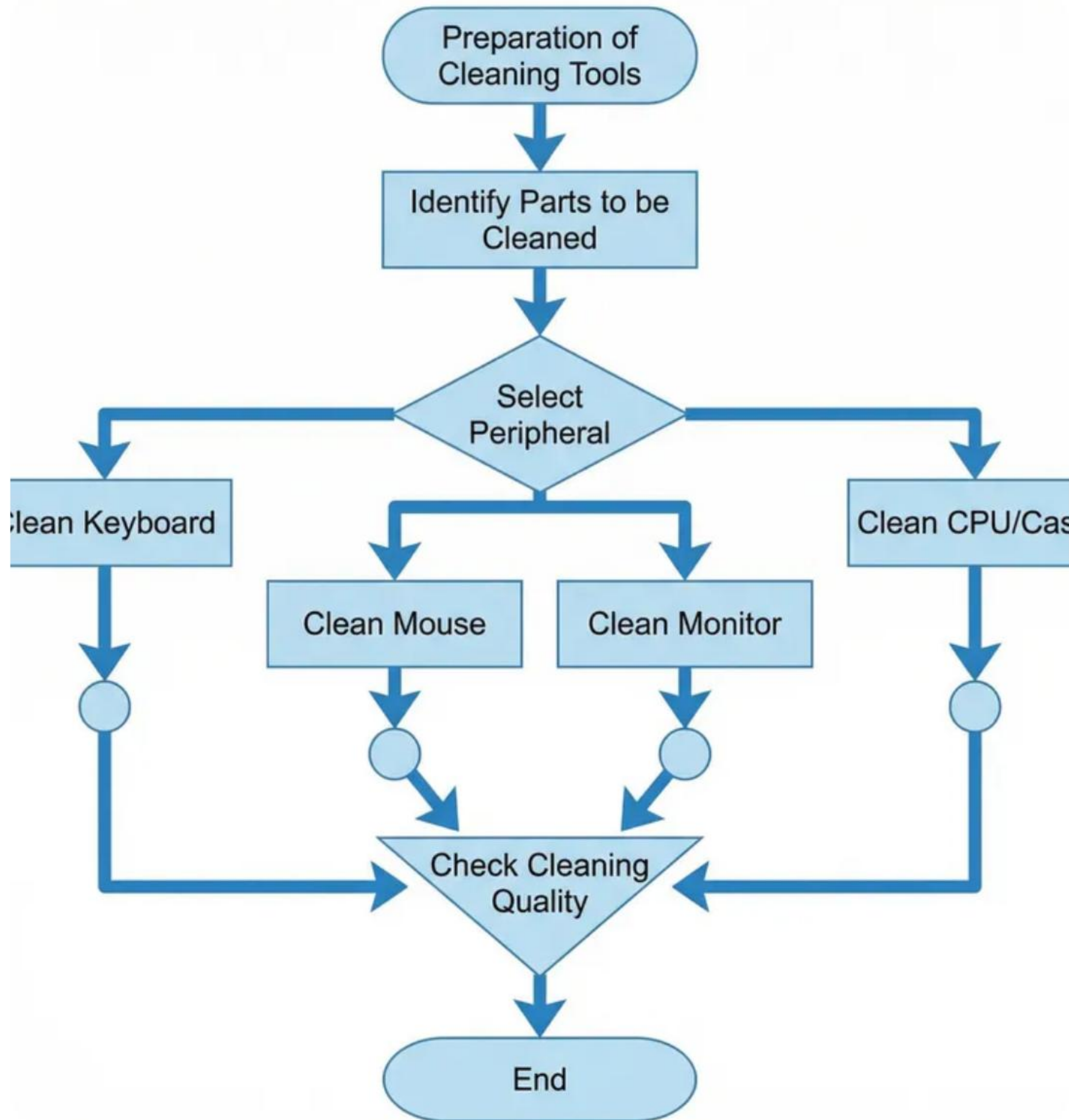


ALGORITHMIC LOGIC

Introduction to Pseudocode

Mastering the pbinfo.ro standard structures



CONCEPT

What is Pseudocode?

- `</>` **Language-Agnostic:** Universal logic blueprint.
- 👁️ **Human Readable:** Focuses on algorithm design.
- 🔒 **Syntax Free:** No strict compiler rules.

Why Learn Pseudocode?



Logic First

Master the algorithm structure.



Easy Translation

Convert to Python or C++.



Standardized

Required for Romanian exams.

Variables & Assignment

Variables store data values in memory. The arrow operator assigns these values.

```
a ← 15 b ← 10 S ← a + b
```

Input and Output

Input

Retrieves user keyboard data.

```
citeste a, b
```

Output

Displays results on screen.

```
scrie S
```

Basic Arithmetic

Operation	Symbol	Example
Addition	+	$a + b$
Subtraction	-	$a - b$
Real Division	/	a / b
Integer Division	[]	$[a / b]$
Modulo (Remainder)	%	$a \% b$

Decisions (IF)

Branching execution paths
based on conditions.

```
daca a > b atunci scrie a altfel scrie b
```

Indefinite Loops

Pre-tested (cattimp)

```
cattimp a <= 10 executa scrie a
```

Evaluates condition before execution.

Post-tested (repeta)

```
repeta scrie a panacand a =
```

Executes at least one time.

The FOR Loop

Fixed iteration count. Automatically increments the variable.

```
pentru i ← 1, 10 executa scrie i
```

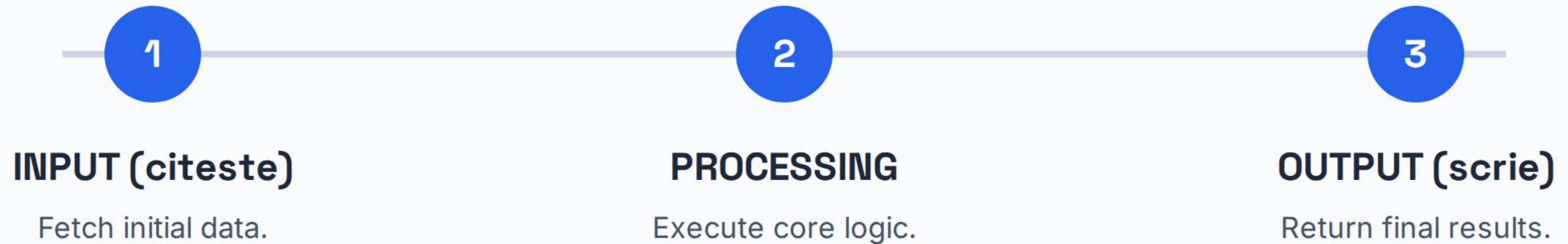


The Taxi Problem

```
cost, dist ← cost / tarif
```

Adapted from Introduction-to-Python.pdf

The IPO Architecture



Ready to Code

Translate your logic into any language.

Image Sources



https://media.easy-peasy.ai/b0df9f99-8afc-42fa-827a-670126ab8725/b40d034b-affd-40a6-a570-82550e97b518_medium.webp

Source: easy-peasy.ai



https://images.stockcake.com/public/b/1/9/b19205ee-536c-42dd-ab31-1470fa5c2d57_large/bustling-city-night-stockcake.jpg

Source: stockcake.com



<http://www.limitlesswalls.com/cdn/shop/files/adobe-177665532.jpg?v=1766068634>

Source: www.limitlesswalls.com